

Block 1 — Research Ideation & Distillation System Specification

1. Purpose

Block 1 is a dual-phase semantic planning system designed for research-driven workflows.

Its purpose is to transform:

- raw voice/text ideation into:
- structured evolving specifications for downstream coding agents and experimentation systems.

The system supports:

- exploratory thinking
- hypothesis generation
- conceptual planning
- ambiguity management
- structured consolidation

The system does NOT perform:

- code execution
- autonomous implementation planning
- orchestration
- scheduling
- direct implementation

The system IS:

a semantic planning and specification engine that sits between research ideation and downstream execution systems.

2. Core Design Philosophy

The architecture separates:

A. Exploration (Creative Mode)

High-entropy thinking:

- uncertainty
- divergence
- competing ideas
- speculative reasoning
- conceptual experimentation

from:

B. Consolidation (Distillation Mode)

Low-entropy structuring:

- requirement extraction
- conceptual stabilisation
- ambiguity reduction
- downstream preparation

This separation is critical.

Premature convergence destroys creativity.

Lack of consolidation prevents execution.

3. High-Level Architecture

```
Voice Input
  ↓
Whisper (local transcription)
  ↓
Transcript Buffer
  ↓
Mode Selector
  ├── Creative Mode
  └── Distillation Mode
  ↓
Semantic Processing Engine (LLM)
  ↓
Living Markdown Specification
```

↓
Human Interaction Loop

4. Core Components

4.1 Whisper Layer

Purpose

Convert spoken input into text.

Notes

- Hosted locally (DGX Spark)
- Independent from Block 1 logic
- Treated purely as an input adapter
- No semantic reasoning occurs here

Output

raw transcript chunk

4.2 Transcript Buffer

Purpose

Maintain ordered conversational and ideation history.

Responsibilities

- append transcript chunks
- preserve chronological interaction history
- provide contextual continuity to semantic engine

Notes

The transcript buffer is external state.

The LLM itself is stateless.

4.3 Living Markdown Specification

Purpose

The canonical structured artefact representing current intent.

This is the primary output of Block 1.

Structure

```
## Objective

## Context

## Ideas

## Hypotheses

## Constraints

## Assumptions

## Requirements

## Conceptual Plan

## Open Questions
```

4.4 Semantic Processing Engine

Purpose

Transform:

- transcript history
- existing specification
- optional contextual grounding

into:

- updated structured semantic state

Critical Architectural Principle

The LLM is NOT the system.

The LLM is:

a semantic transformation function operating over explicit external state.

5. Creative Mode (High Entropy)

Purpose

Enable exploratory research thinking without premature convergence.

Behaviour

Allowed:

- divergent thinking
- multiple competing ideas
- contradictions
- speculative hypotheses
- open-ended questioning
- reframing
- alternative generation
- conceptual expansion

Encouraged:

- uncertainty
 - incomplete thoughts
 - conceptual experimentation
-

Creative Interaction

The system may:

- ask exploratory questions

- suggest multiple alternative directions
- propose hypotheses
- reframe concepts
- expand possibility space

The system MUST NOT:

- force convergence
 - prematurely finalise structure
 - commit to implementation decisions
-

Creative Mode Output Targets

Creative outputs populate:

Ideas

Hypotheses

Open Questions

6. Distillation Mode (Low Entropy)

Purpose

Convert exploratory thinking into stable structured specifications suitable for downstream execution systems.

Behaviour

The system:

- merges transcript history into structured state
 - extracts requirements
 - stabilises constraints
 - resolves contradictions where appropriate
 - reduces ambiguity carefully
-

Distillation Rules

The system MUST:

- preserve stable structure
- perform incremental updates
- maintain semantic consistency
- preserve valid existing information unless contradicted
- explicitly surface unresolved ambiguity

The system MUST NOT:

- invent new goals
 - autonomously select architectures
 - generate executable plans
 - decide implementation details without user approval
-

Distillation Interaction

Allowed:

- clarification questions
- bounded option proposals
- ambiguity resolution requests

Goal:

compress uncertainty into stable structure.

7. Interaction Layer

The system is interactive in both modes.

7.1 Creative Interaction

Optimises for:

- exploration
- expansion
- hypothesis generation

Interaction style:

- open-ended
 - speculative
 - divergent
-

7.2 Distillation Interaction

Optimises for:

- precision
- structure
- convergence

Interaction style:

- constrained
 - targeted
 - ambiguity-reducing
-

8. Mode Switching

8.1 Creative → Distillation

Triggered by phrases such as:

- "distil this"
 - "convert to spec"
 - "finalise"
 - "prepare for coding agent"
-

8.2 Distillation → Creative

Triggered by phrases such as:

- "brainstorm"
- "explore"
- "what are alternatives"

- “expand possibilities”
-

9. State Management

Critical Principle

The API/LLM is stateless.

Conversation continuity is implemented externally.

The system itself owns:

- transcript history
 - specification state
 - mode state
 - optional graph context
-

Recommended State Object

```
{
  "transcript_buffer": [],
  "current_spec_markdown": "",
  "mode": "creative | distillation",
  "codebase_graph": {},
  "metadata": {}
}
```

Mental Model

The system behaves as:

```
next_state = f(previous_state, new_input)
```

The model performs:

- semantic transformation

The system itself performs:

- state persistence
 - interaction management
 - orchestration
-

10. Optional Codebase Graph Injection

Purpose

Provide structural grounding from an existing codebase.

Useful when:

- ideation depends on current architecture
 - proposed changes interact with existing systems
 - planning requires awareness of implementation structure
-

Important Constraint

The graph is:

- optional
- passive
- read-only

It is NOT:

- an autonomous retrieval system
 - an execution planner
 - a mutable knowledge base
-

Usage Rule

Codebase graph injection occurs ONLY:

- when explicitly enabled
 - typically during Distillation Mode
-

Example Graph Schema

```
{
  "modules": [
    {
      "name": "string",
      "summary": "string"
    }
  ],
  "files": [
    {
      "path": "string",
      "summary": "string",
      "functions": ["string"]
    }
  ],
  "edges": [
    {
      "from": "string",
      "to": "string",
      "type": "depends_on | calls | imports"
    }
  ]
}
```

11. Output Contracts

11.1 Creative Mode Output

Outputs:

- exploratory notes
- hypotheses
- questions
- conceptual branches

Stored in:

- Ideas
- Hypotheses
- Open Questions

No structural finalisation occurs.

11.2 Distillation Mode Output

Outputs:

- single updated Markdown specification

No:

- execution logic
 - implementation code
 - orchestration plans
-

12. Failure Handling

12.1 Creative Mode

Ambiguity is allowed and encouraged.

The system should:

- expand uncertainty
 - explore alternatives
 - preserve divergent thinking
-

12.2 Distillation Mode

If ambiguity exists:

- place ambiguity into:
 - Open Questions
 - do NOT hallucinate requirements
 - do NOT infer hidden intent
-

13. System Invariants

The following invariants must always hold:

- The Markdown spec is the single source of truth
 - All updates are incremental unless explicitly reset
 - The LLM is stateless
 - The system owns state externally
 - No autonomous decision-making occurs
 - No execution planning occurs
 - Exploration and consolidation remain separated
-

14. Unit Tests

14.1 Incremental Update Preservation

Input:

- existing spec
- unrelated transcript update

Expected:

- unrelated sections unchanged
-

14.2 No Full Regeneration

Input:

- small requirement modification

Expected:

- localised update only
 - stable structure preserved
-

14.3 Contradiction Handling

Input:

- conflicting objective

Expected:

- explicit overwrite or contradiction note
-

14.4 Ambiguity Handling

Input:

- vague or incomplete statement

Expected:

- ambiguity placed into Open Questions
-

14.5 Creative Divergence Preservation

Input:

- multiple conflicting ideas

Expected:

- all ideas preserved during Creative Mode
-

14.6 Distillation Convergence

Input:

- exploratory transcript + distillation trigger

Expected:

- structured consolidated spec
-

14.7 Grounded Mode Injection

Input:

- codebase graph enabled

Expected:

- graph influences semantic context only
-

14.8 Structural Integrity

Output must always contain:

```
## Objective
## Context
## Constraints
## Requirements
## Conceptual Plan
## Open Questions
```

15. Final Definition

Block 1 is:

a dual-phase semantic planning system that transforms exploratory research thinking into structured executable specifications while preserving separation between creative exploration and downstream consolidation.